



RECOMMENDER SYSTEMS

A Collaborative Filtering Approach Based on Naïve Bayes Classifier

Report 2 — Midterm vs. Final: Summary of Differences

Supervisor: Gamze USLU, Ph.D.

Ömer MANAV – 210209003

Kaan Bahadır SAĞRA – 210201011

Overview

The final submission extends the midterm NBCF implementation with two algorithmic improvements and three evaluation additions. The midterm's temporal hybrid achieved MAE = 0.792 but had a non-functional confidence system (0% coverage) and evaluated only a single narrow user group. The final version addresses both issues and adds RMSE, stratified user-group analysis, and an alpha sensitivity study.

Algorithm Changes

Change	Description	Impact
Top-K Neighbor Selection (NEW)	Candidates ranked by co-rater overlap ($ \text{users}(i) \cap \text{users}(j) $ for UB; $ \text{items}(u) \cap \text{items}(v) $ for IB). Top K=30 retained; rest discarded.	Latency: 34.8ms $\rightarrow$ 14.3ms (−59%) MAE medium users: 0.803 $\rightarrow$ 0.753 (−6.2%) MAE heavy users: 0.740 $\rightarrow$ 0.777 (+slight)
Confidence Score Fixed (UPDATED)	Entropy-based measure replaced with max-probability: $\text{conf} = \max P(y u, i)$ applied to UB-Temporal scores before geometric hybrid mixing.	Coverage: 0% $\rightarrow$ 29.3% (at $\tau=0.45$) Filtered MAE: 0.706 (−11.5% vs overall)

Evaluation Changes

Addition	Midterm	Final
RMSE metric	Not reported	Added alongside MAE
User group eval	20–30 rating users only	Light / Medium / Heavy (300 each)
Alpha sensitivity	$\alpha=0.01$ fixed	5 values: 0.001 – 1.0
Confidence sweep	Single threshold $\tau=0.30$	Sweep $\tau=0.20$ to 0.95 (Figure 8)
New figures/tables	6 figures, 5 tables	9 figures, 7 tables

Metric Comparison

Model	MAE	RMSE	Coverage	Latency
UB	0.861	1.252	—	—
IB	2.335	2.638	—	—
Hybrid	1.102	1.495	—	28.3 ms
Hybrid + Temporal	0.792	1.123	0%	34.8 ms
TopK + Temporal [NEW]	0.798	1.126	29.3%	14.3 ms